

ASSIGNMENT 2

Subject: Software Testing

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Question 1: Test Case Review

a. Explain 2 key benefits of Test Case Review in ensuring quality.

- Improves test case quality and detects defects early in the test design phase that help to reduce execution time.
- Enhances team collaboration and knowledge sharing.

Question 2: Defect Life Cycle and Test Closure

a. Define Test Environment (Test Bed) and explain its importance before Test Execution.

A Test Environment also called 'Test Bed' is the setup of hardware, software, network, database and configurations needed to execute test cases. It is the 'playground' where testers validate the software before release.

Importance:

- Ensures real time testing conditions.
- Helps identify environment related defects.
- Provides accurate test results.

b. Describe the Defect Life Cycle, including terms Mistake, Bug, Defect, and Error.

The 'Defect Life Cycle' or 'Bug Life Cycle' describes the stages a defect goes through from discovery to closure.

Phases of Defect Life Cycle:

1. **New:** Tester finds a defect and logs it in the defect tracking tool.
2. **Assigned:** The defect is assigned to a developer for fixing.
3. **Open:** Developer starts analysing and working on the defect.
4. **Fixed:** Developer fixes the defect and updates the status.
5. **Retest:** Tester tests the application again to check if the defect is fixed.
6. **Reopened:** If the defect still exists after fixing, it is reopened and sent back to the developer.
7. **Closed:** If the defect is fixed successfully, the tester closes it.

Terms in Defect Life Cycle:

- **Mistake:** Human action that produces incorrect result. For example: a developer types = instead of ==.
- **Error:** Incorrect logic in code or program. For example: the program fails a calculation due to wrong logic.
- **Defect:** Deviation from the expected behaviour found during testing. For example: Tester sees incorrect output during testing.
- **Bug:** Informal term for defect. For example: there is a login bug in the app.

c. Differentiate between a Test Summary Report and Test Signoff Document with 1 point each.

- **Test Summary Report:** Document that summarises testing activities, results, and defect status.
- **Test Signoff Document:** Formal approval stating testing is completed and product is ready for release.

Test Case Design and Test Execution

- a. Select a simple application. Describe the objectives and scope of test case design for this application.
- b. Prepare a Test Case Document including: Features to be tested, Entry and Exit Criteria, Preconditions.
- c. Design at least 6 test cases in a table format with fields: Test Case ID, Description, Steps, Test Data, Expected Result.
- d. Create at least 5 executed test cases in table format with the following fields: Test Case ID, Description, Expected Result, Actual Result, Status (Pass/Fail), Comments.

Application Selected: Amazon-Login Page

Objectives:

- To verify that the functionality works correctly.
- To ensure valid users can log in successfully.

- To ensure invalid users are restricted.
- To validate input fields & error messages.

Scope of testing:

- Email ID/ mobile number and password fields.
- Login button functionality.
- Validation messages.
- Successful and unsuccessful login scenarios.
- Basic security validations (empty fields, invalid inputs).

Test Case Document:

Features to be tested:

- Login button.
- User login with valid email ID/ mobile number and password.
- Login attempts with incorrect password.
- Login with empty fields.
- Validation messages for incorrect inputs.
- Successful access to user account after login.

Entry Criteria:

- Amazon application is installed or accessible through browser.
- Login page is available.
- Test data (valid email and password) is ready.

Exit Criteria:

- All planned test cases are executed.
- Expected results are verified.
- Defects are fixed.
- Test summary report is prepared.

Preconditions:

- User should have a registered Amazon account.
- Internet connection should be available.
- Application should load successfully.

Test Case Design and Execution of Amazon-Login Page:

Scenarios	Test Case ID	Description	Steps	Test Data	Expected Result	Actual Result	Status(Pass/Fail)	Comments
Scenario 1: Verify login with valid email and password	TC1	Login with valid email id and password	1. Open Amazon login page		User should successfully login and redirect to home page	As expected	Pass	Working as expected
			2. Enter valid email id	Email id: user@test.com				
			3. Enter correct password	Password: test123				
			4. Click login					
Scenario 2: Verify login with valid mobile number and password	TC2	Login with valid mobile number and password	1. Open login page		User should successfully login and redirect to home page	As expected	Pass	No issue
			2. Enter valid mobile number	Mobile n.o: 987456221				
			3. Enter correct password	Password: test123				
			4. Click login					
Scenario 3: Verify login with incorrect password	TC3	Login with incorrect password	1. Enter valid email id/ mobile n.o	Email id: user@test.com or Mobile n.o: 987456221	Error message should be displayed	As expected	Pass	No issue
			2. Enter wrong password	Password: wrong123				
			3. Click login					
Scenario 4: Verify login with empty field	TC4	Login with empty email/mobile n.o field	1. Leave email/mobile n.o field empty	Email/mobile n.o: Blank	Error message should be displayed	As expected	Pass	No issue
			2. Enter password	Password: test123				
			3. Click login					
Scenario 5: Verify login with empty password field	TC5	Login with empty password field	1. Enter valid email id/ mobile n.o	Email id: user@test.com or Mobile n.o: 987456221	Error message should be displayed	As expected	Pass	No issue
			2. Leave password empty	Password: Blank				
			3. Click login					
Scenario 6: Verify login with invalid email format	TC6	Login with invalid email format	1. Enter invalid email format	Email id: user123	Error message should be displayed	As expected	Pass	No issue
			2. Enter correct password	Password: test123				
			3. Click login					